A 2B Single-Pass Decision Model as a Mobile Agent: Per-Step Accuracy, Task Success, and What the Gap Is Made Of

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Abstract

We turn a 2B *decision model*, LiquidAI's d1-2B, into a mobile agent. At every step the model reads the screenshot, the goal, its own action history and the screen's accessibility elements once. It then answers a fixed set of typed questions in a single forward pass and generates no tokens. The questions are: which action, which element, which scroll direction, which app, and which span of the input to type or report. We fine-tune on AndroidControl with synthetic augmentation. We add 1,736 goal-to-app rows built from AndroidWorld's task generator; they cover 97 of the 116 evaluated templates. No AndroidWorld rollout is used.

Results. On AndroidControl-High the model gets 61.7% of steps right (5,213/8,444), but only 22.3% of episodes have every step right (344/1,543). On AndroidControl-Low the figures are 89.3% of steps and 60.4% of episodes. Under the official AndroidWorld protocol (116 tasks, seed 30, one attempt) it scores 25/115 = 21.7%, at a median of 135 ms per decision.

Analysis. The episode-level AndroidControl rate and the AndroidWorld rate are nearly equal, while the per-step rate suggests much more. We decompose the gap. Of the 91 failed AndroidWorld tasks, 51 ended with a wrong "done" and 40 ran out of steps. About 36 need abilities a single pass under our constraints does not have: arithmetic, dates, memory across screens, and text that exists only in pixels. Harness and environment fixes alone moved the score from 15.5 to 25 on a fixed checkpoint. We report negative results with the positive ones.

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1. Introduction

Published mobile GUI agents are almost all *generative*. A vision-language model writes a thought and an action string at each step, often over several calls (planner, actor, reflector). A *decision model* works another way. It reads the state once and returns probabilities over options that the caller supplies, and it writes no free text (TypeSafe AI, 2026a; Liquid AI, 2026). For an on-device agent this offers three things: low latency, an output schema that cannot be broken, and a confidence for each answer. Whether such a model can *act*, and not only classify, is an open question.

We study that question with measurements, not with a leaderboard claim. Our contributions:

1. A **decision contract** for phone control: a text-and-image state, typed questions, and outputs that copy spans from the input. One function renders it for training, evaluation and serving.
2. Results on AndroidControl (Low and High) in the standard Type / Grounding / SR format, and on AndroidWorld under its official protocol, each with its denominator.
3. A breakdown of the gap between per-step accuracy and task success, based on a trace-level reading of all 91 failed AndroidWorld tasks.
4. A record of what did not work, and of harness and environment defects that changed scores without any change to the model.

2. Related Work

Generative mobile agents. M3A, the reference agent of AndroidWorld, keeps a per-step LLM summary as memory and reports 30.6% with GPT-4 Turbo (Rawles et al., 2024). UI-TARS (Qin et al., 2025), MAI-UI (Zhou et al., 2025), GUI-Owl and Mobile-Agent-v3 (Ye et al., 2025), and UI-Venus-1.5 (Venus Team et al., 2026) are end-to-end VLMs that generate text. MobileRL adds online reinforcement learning on generated AndroidWorld task variants (Xu et al., 2025). V-Droid scores candidate actions

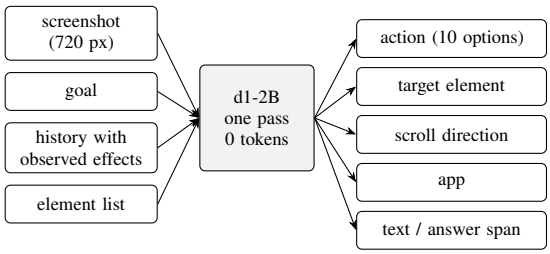


Figure 1. The decision contract. One forward pass answers five typed questions; the harness maps the answers to one AndroidWorld action.

with a verifier and trains on AndroidWorld apps and tasks (Dai et al., 2025). AndroidGen collects its trajectories in the AndroidWorld emulator and grows a retrieval database during evaluation (Lai et al., 2025). Table 1 lists their AndroidWorld numbers. These numbers are context, not a ranking: model size, calls per step, generated reasoning, extra components and benchmark-derived training data all differ, and none of them was produced by our harness.

Decision models. TypeSafe's Jev and LiquidAI's d1 answer typed questions (choice, yes/no, score, span, box) in one pass (TypeSafe AI, 2026a; Liquid AI, 2026). TypeSafe's guidance says such models should "keep math in code, which includes counting and date comparison" (TypeSafe AI, 2026b). Our failure analysis reaches the same point from the agent side (Section 5.8).

Data and per-step benchmarks. AndroidControl gives a goal (High) and a per-step instruction (Low) for 15,283 demonstrations (Li et al., 2024). Its usual scorer compares each predicted step with the demonstrator's step, given the demonstrator's history. AndroidControl-Curated cleans the test set and reports a per-step success rate (Leung et al., 2025). AMEX adds element-level annotations and also reports only per-step metrics (Chai et al., 2024). None of these checks whether a task was completed.

3. The Decision Contract

Figure 1 shows the contract.

State. The input at each step is the screenshot plus a text state. The text holds the goal, a progress line, the current app, and the previous actions, each with the effect the harness observed after it ("screen changed" or "no change"). It ends with the element list rendered from the accessibility tree. One function renders this state for training, evaluation and serving, so the three cannot drift apart.

Questions. Each output is a typed choice: the action, from {click, type, long press, scroll, wait, back, home, open app, done, answer}; the target element; the scroll direction; and the app. Typed text and answers are *pointer reads*, that is, spans copied from the text state. In d1, span questions read positions of the text input only. The model therefore cannot produce text that appears only in the pixels (Section 5.8).

Harness. The harness follows AndroidWorld's official protocol: seeded task instances, partial credit, termination decided by the agent, and no oracle stop. Every episode header records the agent policy that produced it.

4. Training

Schedule. We fine-tune every parameter of d1-2B except the tied embedding and output head, which stay frozen. The vision encoder is trained too: all 132 vision tensors differ from the base checkpoint. Training is one epoch of 2,482 steps with 16 rows per step and a learning rate of 10^{-5}.

Data. The main set has 37,977 rows from AndroidControl's train split. They are rebuilt from the original files, restricted to task kinds similar to AndroidWorld's, and exclude every test episode. We augment them in five ways:

- history lines labelled with the observed effect of each action;
- "no change" rows, which teach the model not to repeat an action that had no effect;
- rows that mark actions to avoid;
- the AndroidControl-Low step instruction, shown on a fixed half of the rows;
- episodes that end in an answer.

Disclosure. A further 1,736 rows map goal text from AndroidWorld's own task generator to the app to open. They cover 97 of the 116 evaluated templates; 19 templates are held out. No AndroidWorld rollout, screen or checker is used. The rows still expose the model to AndroidWorld's goal phrasing and app names for most tasks. We do not report a split of success between covered and held-out templates.

5. Experiments

5.1. Setup and Metrics

AndroidControl. We use the 8,444-step test split (1,543 episodes) and InfiGUI's scorer without changes. *Type* is action-type accuracy. *Grounding* is the share of click and long-press steps, among those with the right type, whose point lies in the target element. *SR* is the share of steps where the whole action is right. We also report the share of episodes in which every step is right. AndroidControl-Low shows the step instruction; AndroidControl-High shows

Table 1. Published AndroidWorld success rates, for context only (not a ranking). Each value is the paper's own report on the official task suite. "–" means we did not find the field in the source. The AndroidGen percentages fit 111 tasks rather than 116 (our derivation).

Agent / model	Params	Calls	Text	AndroidWorld-derived training data	Extra components	SR (%)
Ours: d1-2B	2B	1	no	goal-to-app rows from the task generator (97/116 templates)	none	**21.7**
M3A, GPT-4 Turbo (Rawles et al., 2024)	–	≥2	yes	no	step summary as memory	30.6
UI-TARS (Qin et al., 2025)	–	1	yes	not stated	–	46.6
AndroidGen, GPT-4o (Lai et al., 2025)	–	several	yes	trajectories collected in AndroidWorld	planner, retrieval, checks	46.8
MAI-UI-2B (Zhou et al., 2025)	2B	1	yes	not stated	none	49.1
UI-Venus-1.5-2B (Venus Team et al., 2026)	2B	1	yes	not stated	–	55.6
V-Droid (Dai et al., 2025)	8B + LLM	several	yes	AndroidWorld apps and tasks	verifier	59.5
GUI-Owl-7B (Ye et al., 2025)	7B	1	yes	not stated	–	66.4
Mobile-Agent-v3 (Ye et al., 2025)	32B	several	yes	not stated	manager, worker, reflector, notetaker	73.3
MAI-UI-235B-A22B (Zhou et al., 2025)	235B	1	yes	not stated	none	76.7
MobileRL-9B (Xu et al., 2025)	9B	1	yes	online RL on generated task variants	–	80.2

only the goal. Both feed the demonstrator's history at each step.

AndroidWorld. We run 116 tasks with seed 30, one attempt per task (pass@1), and the official checker. SR is the sum of task scores over scored tasks, so a partial credit counts as half. Decoding is argmax, but the environment is not deterministic: in an earlier check, 6 of 20 same-host replays diverged.

5.2. AndroidControl

Table 2 gives AndroidControl results in the format used by recent papers. The other rows are copied from MobileRL's Table 2 (Xu et al., 2025). Those models use other scorers, other action spaces and other history handling: MobileRL, for example, keeps the agent's own history in AndroidControl-High and replaces wrong actions. The rows are therefore not strictly comparable with ours. On Low, our SR (89.3) is 1.5–6.6 points below every listed row. On High the gap is larger: 9–16 points on Type and 10–16 points on SR. High grounding (78.4) is between the MobileRL rows (71.5–78.1) and the UI-TARS and UI-Genie rows (80.5–86.3). Our model has 2B parameters and generates no tokens; the listed models have 7B–72B and generate text. Fine-tuning moved High SR from 40.2 (the same model and questions, not fine-tuned) to 61.7, and whole-episode accuracy from 9.4% to 22.3%.

5.3. Per-Step Accuracy Is Not Task Success

The AndroidControl-High scorer gives the demonstrator's history and screen at every step and never checks the goal, so errors cannot compound. Whole-episode accuracy shows the compounding that the per-step rate hides (Table 3): 59% of one-step episodes are fully right, but only 5% of episodes with ten or more steps.

The whole-episode rate on AndroidControl-High (22.3%) is close to the AndroidWorld rate (21.7%), but the two measure different things. AndroidWorld is harder because errors compound, the agent must stop on its own, and 13 of its 20 apps have no training goals. It is easier because any path that reaches the goal state is accepted. In our runs the two effects roughly cancel, which makes the episode-level rate the better offline proxy.

Termination explains a large share of the gap. The AndroidControl test set has *no* step where "done" is correct, yet the model predicts "done" on 1,171 of 8,444 steps. Of these, 460 are on an episode's last real step and 566 are on steps whose correct action is a scroll. The two groups share 186 steps, so they must not be added.

5.4. AndroidWorld

The model scores 25/115 = 21.7% (24 full successes and two half credits). One task, OsmAndMarker, failed at environment setup and is excluded; the cause is fixed (Section 5.5). For reference, on an earlier harness an agent that declares "done" at step 0 scores 7/113 = 6.6%, because some tasks start already satisfied.

Table 4 breaks the score down by the upstream difficulty labels (61 easy, 36 medium, 19 hard) and Table 5 by complexity level, defined as in MobileRL (the ceiling of the

Table 2. AndroidControl test set (8,444 steps). Type = action-type accuracy; Grounding = target accuracy on click and long-press steps with the right type; SR = step success rate. Rows above the line are copied from MobileRL, Table 2 (Xu et al., 2025) (scorers and protocols differ; not strictly comparable). Our rows use InfiGUI's scorer.

			AndroidControl-Low			AndroidControl-High		
Model	Params	Tokens	Type	Grounding	SR	Type	Grounding	SR
UI-TARS-7B	7B	yes	98.0	89.3	90.8	83.7	80.5	72.5
UI-TARS-72B	72B	yes	98.1	90.8	91.3	85.2	81.5	74.7
UI-Genie-7B	7B	yes	98.1	89.3	94.3	83.5	82.9	74.5
UI-Genie-72B	72B	yes	98.3	95.4	94.8	84.9	86.3	77.0
MobileRL-7B w/o RL	7B	yes	96.7	93.7	91.9	87.0	73.6	72.8
MobileRL-7B w/ RL	7B	yes	96.9	92.4	91.4	86.3	71.5	71.3
MobileRL-9B w/o RL	9B	yes	98.6	95.8	95.9	89.8	78.1	77.5
MobileRL-9B w/ RL	9B	yes	97.9	94.6	94.3	87.0	74.5	73.9
d1-2B, not fine-tuned	2B	none	*pending*	*pending*	*pending*	63.6	63.1	40.2
d1-2B decision model (ours)	2B	none	**96.6**	**90.1**	**89.3**	**74.3**	**78.4**	**61.7**

Our counts: Low — Type 8,160/8,444, Grounding 4,454/4,941, SR 7,544/8,444; High — Type 6,270/8,444, Grounding 3,337/4,257, SR 5,213/8,444. Episodes with every step right: Low 932/1,543 (60.4%), High 344/1,543 (22.3%). Not fine-tuned d1-2B, High: Type 5,371/8,444, Grounding 2,222/3,523, SR 3,391/8,444; episodes 145/1,543 (9.4%).

Table 3. AndroidControl-High: episodes with every step right, by episode length.

Steps	1	2	3	4	5	6	7	8	9	≥10
Right	69	37	70	60	49	19	18	11	4	7
Total	116	122	200	221	256	173	138	102	65	150
%	59	30	35	27	19	11	13	11	6	5

Table 4. AndroidWorld by difficulty (pass@1). Steps are averages per task; "succ." and "fail." split them by outcome; "opt." is the upstream optimal step count.

Difficulty	n	Score	SR (%)	Steps	succ.	fail.	opt.
easy	61	21.0	34.4	8.5	5.5	10.1	5.0
medium	36	3.0	8.3	13.0	6.7	13.6	9.0
hard	18	1.0	5.6	19.3	–	19.3	18.5
all	115	25.0	21.7	11.6	5.7	13.2	8.4

task's complexity value). The "optimal" column averages the upstream optimal step count over all scored tasks in a row. On the tasks the model solved, it used 5.5 steps on easy tasks where the optimum for those tasks is 3.2, and 6.7 steps on medium tasks against 5.0, so its successes are not on the shortest path. Our level counts (45/41/9/21) differ from those MobileRL shows for the same definition (49/41/11/15, read from its figure).

5.5. The Harness and the Environment Move the Score

Table 6 holds the checkpoint and the seed fixed and changes only the harness and the emulator. The score rose from 15.5 to 25 out of 115. With paired per-task comparisons, the two steps gave 5 wins and 0 losses ($p = 0.062$, sign test) and 5 wins and 1 loss ($p = 0.22$).

We found four environment defects by reading traces; each produces failures that look like model failures.

- **Markor** opened on its first-run screens in 6 of 6 Markor episodes on one emulator.
- **OsmAnd**: a database query run as root on a missing database left an empty, root-owned marker database. OsmAnd could not open it, so the OsmAndMarker task failed setup in 3 of 4 runs.
- **VLC**: the media library stays at "Loading" on every emulator. The log shows the library rejecting a folder after the task's own setup clears its media table as root. This cause is inferred, not proven. The Browse tab still reaches the files.
- **Accessibility**: running `uiautomator dump` stops AndroidWorld's accessibility forwarder.

Table 5. AndroidWorld by complexity level (ceiling of task complexity, as in MobileRL).

Level	C1	C2	C3	C4
Tasks	45	41	9	21
Score / scored	19/45	5.5/40	0/9	0.5/21
SR (%)	42.2	13.8	0.0	2.4

5.6. The Model Uses the Image

On the 827-step validation split we replaced every screenshot with a uniform grey image of the same size (Table 7). Step accuracy fell from 519 to 337. Most of the loss is in *which kind* of action to take: action-type accuracy fell by 29 points, and typing and scrolling nearly disappeared. We read this as the model looking for a keyboard or a long list, but that is our interpretation, not a measurement. Element

Table 6. Same checkpoint, same seed; only the harness and the environment change.

Harness	Score	Change (paired result vs. the row above)
A	15.5/115	–
B	20.5/115	answers now reach the checker (every answer had scored 0); a fresh observation when the accessibility tree lags the pixels (cap 8 s). 5 wins / 0 losses.
C	**25/115**	fresh wait also on page change (cap 20 s); a working SIM; Chrome first-run screens cleared before the suite. 5 wins / 1 loss.
D	*pending*	numeric answer format; Back closes the keyboard after typing; Markor and OsmAnd state repaired. Only the 39 affected tasks are re-run.

Table 7. Validation split (827 steps) with real vs. uniform grey screenshots.

	Real	Grey
Step accuracy	519/827	337/827
Action-type accuracy	80.0%	50.8%
Element, given right type	271/414	124/207
Type-text steps right	37/53	3/53
Scroll steps right	40/86	7/86

accuracy given the right type fell less; note that it is measured on different subsets in the two conditions.

5.7. Latency

Over 1,335 AndroidWorld steps, one decision takes 135 ms at the median and 181 ms at the 95th percentile. This is one forward pass on a shared 24 GB GPU, served over HTTP. The environment step takes 1,624 ms at the median, so the device, not the model, dominates the agent loop.

5.8. What the 91 Failed Tasks Are Made Of

We split the failed tasks into four groups. One LLM agent per group, with no shared context, found the first wrong step of each task from its trace and screenshots and checked the upstream success condition. Table 8 summarises the result.

By outcome, 51 of the 91 failures ended with a wrong "done" (49 scored zero and 2 received half credit) and 40 ran out of step budget. From the group reports we estimate that about 36 failures are blocked by our constraints of one model, no planner and AndroidControl-only data. This is an estimate, not a per-task tally. The blocked abilities are:

- arithmetic and date computation;
- memory across screens (counting, duplicates, transcription);
- text that exists only in pixels;
- colour or icon meaning with no training labels;
- information the accessibility tree does not carry (for example screen brightness), and inputs the data cannot teach (clipboard chips, appending to a field).

TypeSafe's own guidance assigns this kind of work to code, not to the model (TypeSafe AI, 2026b).

5.9. Negative Results

Table 9 lists what did not work. Each row is one run unless stated.

5.10. Zero-Shot Pattern Probes

We asked the trained model two extra questions it was never trained on (Table 10). Re-ranking the top five elements with one yes/no question per candidate made element choice worse, although the gold element is in the top five on about 90% of rows. Asking "did the last action move toward the goal?" on the next screen separated good from bad transitions with an AUROC of 0.76. This rests on only 26 negatives.

6. Limitations

- **Power.** One seed and one attempt per task. A difference of about five tasks on AndroidWorld is within noise (sign test $p \approx 0.2$ at 5 wins / 1 loss).
- **Benchmark-derived data.** The goal-to-app rows use AndroidWorld's goal templates.
- **Pending.** The not-fine-tuned baseline on AndroidControl-Low, the latest harness, and a retrained model.
- **No same-harness generative baseline.** We did not run a generative 2B model on our harness, so the published numbers in Tables 1 and 2 are context, not comparisons.
- **Taxonomy rigour.** The failure taxonomy comes from LLM agents reading traces, one per group, without a second reader or an agreement measure.

7. Conclusion

A 2B decision model can operate a phone end to end at 135 ms per decision. On AndroidControl its per-step accuracy overstates its task success by about a factor of three; its

Table 8. First wrong step in the 91 failed AndroidWorld tasks, by group. Counts are tasks.

Group (tasks)	Main causes
Information retrieval (21)	answer visible but never given (4); answer format (1); wrong navigation (4); date, time or colour reasoning (4); counting across screens (2); answer not expressible in the contract (6)
Easy (27)	long press needed (5); *Verify* tasks already satisfied (3); premature done (4); wrong element (4); target scrolled off screen (3); typing replaces instead of appending (2); clipboard chip not in the element list (2); environment or app knowledge (4)
Medium/hard (34)	app procedure (7); values that must be computed, such as dates and end times (6); stops after the first of several items (4); environment (3); form fields not filled (3); duplicates across screens (3); long note content (2); VLC (2); single cases (4)
Visual (9)	navigation with no reading needed (2); drawing or maze pages (2); text only in pixels (4); arithmetic (1)

Table 9. Negative results.

Attempt	Result
Threshold on p(done)	harmful at every threshold on an earlier offline split
Separate "task done?" head	redundant: its label equals "action = done" on 137/137 rows
Masking an action already marked "no change"	2 wins / 2 losses on AndroidWorld; premature stops rose
Weight merge 0.3/0.7 with an earlier model	6 wins / 9 losses
Recovery ("go back") rows	300-step pilot: validation step accuracy 65.2% → 57.7%; back predicted where gold is not back 1.9% → 10.0%
"Already satisfied → done" rows	at 10% of training, false done 108/690 with the rows vs. 67/690 without; a third run without them gave 106/690, so the effect is not attributable
Early stopping rule at 10% of training	invalid: on an earlier run, mid-training metrics did not predict final ones

Table 10. Zero-shot probes. Val = validation (827 rows, 430 with a target element); AC = AndroidControl sample (2,012 rows, 1,224 with a target).

Probe	Val	AC
Element by trained choice	286/430	821/1,224
Top-5 re-rank by yes/no	261/430	772/1,224
Gold element in top 5	388/430	1,084/1,224
Progress question, AUROC	0.68	0.76
negatives / positives	10 / –	26 / 1,550

episode-level accuracy and its AndroidWorld success are both near 22%. The gap has three parts: termination (51 of 91 failures end in a wrong "done"), harness and environment defects that are worth as much as model changes, and abilities a single pass should not be asked to provide (arithmetic, dates, memory and pixel-only text). In line with TypeSafe's guidance, the next steps keep the model for judgments and move computation into code: (1) verify progress on the next screen, which already shows signal without training; (2) train per-candidate re-ranking; (3) answer by map–reduce, with the model judging each row and code counting, summing and formatting.

References

Chai, Y., Huang, S., Niu, Y., et al. AMEX: Android multi-annotation expo dataset for mobile GUI agents. *arXiv preprint arXiv:2407.17490*, 2024.

Dai, G., Jiang, S., Cao, T., et al. Advancing mobile GUI agents: A verifier-driven approach to practical deployment. *arXiv preprint arXiv:2503.15937*, 2025.

Lai, H., Gao, J., Liu, X., et al. AndroidGen: Building an android language agent under data scarcity. *arXiv preprint arXiv:2504.19298*, 2025.

Leung, H. F., Xi, X., and Zuo, F. AndroidControl-Curated: Revealing the true potential of GUI agents through benchmark purification. *arXiv preprint arXiv:2510.18488*, 2025.

Li, W., Bishop, W., Li, A., et al. On the effects of data scale on UI control agents. *arXiv preprint arXiv:2406.03679*, 2024.

Liquid AI. Open d1. https://www.liquid.ai/blog/d1-open, 2026.

Qin, Y., Ye, Y., Fang, J., et al. UI-TARS: Pioneering automated GUI interaction with native agents. *arXiv preprint arXiv:2501.12326*, 2025.

Rawles, C., Clinckemaillie, S., Chang, Y., et al. AndroidWorld: A dynamic benchmarking environment for autonomous agents. *arXiv preprint arXiv:2405.14573*, 2024.

TypeSafe AI. Introducing system one models and Jev. https://typesafe.ai/blog/introducing-system-one-models-and-jev, 2026a.

TypeSafe AI. Jev 1.13: Model jaggedness. https://docs.typesafe.ai/model-jaggedness/jev-1.13, 2026b.

Venus Team, Gao, C., Gu, Z., et al. UI-Venus-1.5 technical report. *arXiv preprint arXiv:2602.09082*, 2026.

Xu, Y., Liu, X., Liu, X., et al. MobileRL: Online agentic reinforcement learning for mobile GUI agents. *arXiv preprint arXiv:2509.18119*, 2025.

Ye, J., Zhang, X., Xu, H., et al. Mobile-Agent-v3: Fundamental agents for GUI automation. *arXiv preprint arXiv:2508.15144*, 2025.

Zhou, H., Zhang, X., Tong, P., et al. MAI-UI technical report: Real-world centric foundation GUI agents. *arXiv preprint arXiv:2512.22047*, 2025.

A. Artifacts

Every number in the paper is recomputed from files in the project repository.

- AndroidControl per-step predictions (High and Low): `benchmarks/results/2026-11-12-v4/actest/perstep-*.jsonl`.
- AndroidWorld traces: `benchmarks/results/2026-11-12-maskenv/` and `benchmarks/results/2026-11-12-armsAB/`; tables by difficulty and level: `benchmarks/aw/aw_report.py`.
- Failure analysis: `benchmarks/results/2026-11-12-failanalysis/`.
- Image ablation and changed weights: `benchmarks/results/2026-11-12-imgablate/`.
- Pattern probes: `benchmarks/results/2026-11-12-patterns/`.
- AndroidWorld task metadata: `references/raw/androidworld-task_metadata.json`.